

Laxmi Charitable Trust's
Sheth L.U.J College of Arts & Sir M.V. College of Science and Commerce
Department of Information Technology (Bsc.IT Semester III)
Python for Data Science Practical

Practical - II

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Class :Sy-It	Batch : 02
Date of Assignment :22/07/26	Date/Time of Submission :22/07/26

Questions:

2a. Write a python code to create nested tuples.

Code:

```
stud=("Mehek",11, ("BSC.IT","Sem 3"),85)
print("Name= ", stud[0])
print("Roll No= ", stud[1])
print("Department= ", stud[2][0])
print("Semester= ", stud[2][1])
print("Percentage= ", stud[3])
```

Output:

```
-----Nested Tuples -----
Name=  Mehek
Roll No=  11
Department=  BSC.IT
Semester=  Sem 3
Percentage=  85
```

2b. Write a python code to sort the nested tuple using the sorted () function.

Code:

```
data= [(1,3),(7,4),(9,6),(5,2)]
```

```
sorted_data= sorted(data, key=lambda x: x[1])
print(sorted_data)
```

Output:

```
-----Sorted Data -----
[(5, 2), (1, 3), (7, 4), (9, 6)]
```

2c. Write a python code to copy or clone a list.

Code:

```
mehek=[(1,3),(7,4),(9,6)]
list1=mehek
print(list1)
```

Output:

```
-----Clone list -----
[(1, 3), (7, 4), (9, 6)]
```

2d. Write a python code to check the immutability property of python tuples.

Code:

```
print("-----immutability -----")
fruits=("mango","apple","orange","kiwi")
print(fruits)
fruits[2]="Dragon fruit"
print(fruits)
```

Output:

```
-----immutability -----
('mango', 'apple', 'orange', 'kiwi')
Traceback (most recent call last):
  File "C:\Users\DELL\OneDrive\Desktop\Mehek\SEM 3\PDS
\PDS2.py", line 24, in <module>
    fruits[2]="Dragon fruit"
TypeError: 'tuple' object does not support item assign
ment
```